

QuantEdge Realtime Trading Intelligence Platform Detailed Product Requirements Document

1. Product Overview

QuantEdge is a personal realtime trading intelligence platform designed to help identify high-probability intraday trading opportunities, monitor active positions, project short-term price movement ranges, and determine optimal selling zones using multiple predictive engines.

The application is not intended to function as a guaranteed-profit trading bot. Instead, it acts as a realtime analytical system that continuously studies market behavior, evaluates probabilities, projects likely short-term candle movement, and assists in decision-making during active trades.

The platform is designed as a private single-user system with no authentication, no user management, and no SaaS complexity. The application opens directly into the dashboard and focuses entirely on realtime analytics and market intelligence.

2. Product Vision

The primary goal of QuantEdge is to reduce poor trading decisions caused by emotional reactions, impulsive entries, delayed exits, and lack of market clarity.

The platform continuously attempts to answer three important questions:

1. Which stock or option currently has the strongest probability of movement?
2. If a position is active, what is the likely short-term movement in the next 1, 5, or 10 minutes?
3. Based on momentum and projected movement, when is the best time to sell or exit?

The platform should feel analytical, calm, intelligent, and technically serious rather than behaving like a gambling-style trading application.

3. Market Analysis Workflow

When the market opens, the platform immediately begins collecting realtime market data from NIFTY, BANKNIFTY, FINNIFTY, NSE stocks, and selected options contracts.

The system continuously analyzes momentum, volatility, candle structure, trend continuation, reversal pressure, sector strength, volume activity, and option behavior.

The platform ranks and filters the strongest opportunities currently available in the market and presents them with confidence levels, expected movement ranges, and risk analysis.

4. Trade Opportunity Engine

The dashboard displays a ranked list of realtime opportunities. Each recommendation contains:

- Stock or option name
- Market direction
- Confidence score
- Expected movement range
- Risk level
- Required capital
- Suggested stop loss
- Suggested exit zone
- Reasoning behind the signal

The purpose is not simply to say BUY or SELL. The purpose is to explain why the opportunity exists, how strong it is, and how risky it currently is.

5. Live Position Intelligence

Once a trade is entered manually, the user inputs invested amount, entry price, and quantity or lot count.

The platform then activates realtime monitoring for that position and continuously analyzes:

- Momentum strength
- Volume pressure
- Candle structure
- Trend exhaustion
- Reversal probability
- Volatility behavior

The system generates projected short-term movement ranges for the next 1 minute, 5 minutes, and 10 minutes.

6. Multi-Model Prediction System

The platform uses multiple prediction engines simultaneously instead of relying on a single forecasting method.

Prediction engines include:

- Momentum Engine
- Volatility Engine
- Mean Reversion Engine
- Machine Learning Engine
- Monte Carlo Simulation Engine

Each engine independently produces movement projections and confidence ranges. The system compares model performance and validates which engine performs best under different market conditions.

7. Candle Projection Visualization

The chart interface contains realtime candles along with projected future candles.

Projected candles are displayed using semi-transparent and dashed visual styles to clearly separate predicted movement from actual market movement.

Multiple future paths from different prediction engines are displayed simultaneously with confidence percentages and projected movement ranges.

8. Exit Intelligence System

The platform continuously determines whether:

- Momentum is weakening
- Reversal risk is increasing
- Volatility is expanding
- Profit booking is likely

Based on this analysis, the system dynamically suggests:

- Hold
- Partial Exit
- Aggressive Exit
- Full Sell

The system continuously attempts to help determine the safest and most efficient selling zones.

9. Budget-Based Filtering

The platform allows manual capital constraints. The user can define:

- Total available capital
- Maximum affordable premium
- Preferred lot count
- Maximum acceptable risk

The system automatically avoids oversized positions and filters only realistic opportunities that match the available budget.

10. Accuracy Validation System

After market close, the platform evaluates all predictions against actual market behavior.

The system measures:

- Directional accuracy
- Range accuracy
- Prediction deviation
- Confidence calibration
- Exit timing accuracy

The goal is to continuously improve the forecasting quality and identify which prediction engines perform best under specific market conditions.

11. Adaptive Intelligence

The platform learns which models perform best during:

- Trending markets
- Sideways markets
- High volatility sessions
- Low liquidity conditions

Confidence weighting dynamically adjusts depending on current market behavior and recent model performance.

12. UI/UX Design Requirements

The interface should feel:

- Minimal
- Calm
- Analytical
- Premium
- Modern
- Realtime

The UI should avoid noisy trading aesthetics, flashing elements, and gambling-style design patterns.

Design language:

- White background
 - Graphite and light black accents
 - Muted green for positive movement
 - Muted red for negative movement
 - Spacious layout with large whitespace
 - Thin borders and lightweight cards
 - Smooth realtime transitions and subtle motion
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13. Technical Architecture

Frontend Stack:

- Next.js
- TypeScript
- TailwindCSS
- Framer Motion
- TradingView Lightweight Charts

Backend Stack:

- Next.js API Routes
- TypeScript
- WebSockets

Data Layer:

Supabase may be used for:

- Prediction history
- Historical validations
- Trade archives

- Analytics storage

Forecasting Layer:

A separate Python forecasting service handles:

- Statistical calculations
- Simulations
- ML projections
- Forecast engines

Recommended Python stack:

- FastAPI
 - pandas
 - numpy
 - scikit-learn
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14. MVP Scope

The first version should focus only on:

- Live market scanner
- Opportunity ranking
- Realtime projections
- Candle forecasting
- Active trade tracking
- Sell intelligence
- Accuracy validation

The first version should avoid:

- Automated trading
- Broker execution
- Social features
- Portfolio systems
- Mobile apps

The immediate priority is building reliable realtime market intelligence.
